

NICHOLAS HO YUN HEE

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Education

National University of Singapore

Bachelor of Engineering, Electrical Engineering

- Second major in Mathematics
- Relevant classes: Programming, Multivariable Calculus, Robotics, Linear Algebra, Statistics

Experience

Ecovolt, Software Engineering Intern

Jan – May 2026

- Lead in setting up an IoT testbed to validate endpoint whitelisting, supporting secure and reliable device communication, documented testbed architecture, setup procedures, and testing for future development.
- Supporting development of a smart water meter product, assisting with system integration.
- Spearheaded creating new functions for water meter via cpp, creating abstraction layer for software engineering team integrate with rest of company's data monitoring software

Orc4bikes Exco @ Residential College 4, Head

Jan 2026 – Present

- Coordinate with team ensure operations of the interest groups, including hosting sessions, maintaining of equipment and marketing; manage cashflow from renting out bicycles
- Manage automated system with bots that facilitates the process of renting bicycles to residents

Private Mathematics Tutoring, Tutor

Jan 2025 - Present

- Taught mathematics to pupils at the primary, secondary, and high school level.
- Learnt to explain advanced, complicated concepts in simpler forms so pupils can easily grasp

Awards & Scholarship

- NUS ASEAN Merit Undergraduate Scholarship Recipient - Full ride scholarship award
- MOE (Singapore) ASEAN Pre-University Scholarship Recipient

Projects

GPU Transformer Inference Optimization (Python, PyTorch, Triton, CUDA)

- Achieved a $7.45\times$ geometric-mean speedup over an explicit PyTorch reference across 13 input configurations on an RTX 3060 Laptop GPU, with all cases passing elementwise accuracy checks
- Developed fused Triton kernels for residual addition, layer normalization, and feed-forward operations; combined mixed precision, packed QKV projections, and CUDA Graph replay to reduce memory traffic and kernel-launch overhead
- Tuned batch chunking to process 10,000 sequences within 3.47 GiB reserved GPU memory; validated numerical accuracy using seeded inputs and measured median latency with CUDA events

Volatility-Targeted Crypto Momentum Backtest (Python, Pandas, NumPy, yfinance)

- Built a time-series momentum backtester for Bitcoin, Ethereum, and Solana using volatility-based position sizing, lagged trading signals, and turnover-based transaction costs of 0.6%.
- Implemented rolling walk-forward optimization across six momentum lookbacks using 365-day training windows and 90-day out-of-sample periods; evaluated CAGR, Sharpe ratio, volatility, and maximum drawdown against an equal-weighted benchmark.

Skills

- Programming: Python, C/C++
- Tools & Libraries: NumPy, pandas, PyTorch, Triton, CUDA Graphs, Matplotlib, Jupyter Notebook, GitHub
- Languages: English (Native), Chinese (Native), Malay (Basic), Cantonese (Basic)

Co-Curricular Activities

- Jane Street's Estimathon® @ NUS 2025: Made educated guesses via critical deduction, general knowledge and basic statistics; clinched 5th place with a team of peers
- Competed in IMC's trading simulation; built an arbitrage strategy using an automated python bot, where it traded in a simulated market and compared bid-ask prices of basket of goods compared to components
- Participated in Hack4Good 2026, developing a Telegram bot to streamline event registration and provide organisers with easy access to participant information.